

Sameer Aryal

CONTACT INFORMATION

Stanley Center for Psychiatric Research
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RESEARCH INTERESTS

Integrative neurobiology of psychiatric and neurodevelopmental disorders; functional analysis of disease-associated genetic variation; synaptic biology; transcriptomics, proteomics, and single-cell genomics; computational integration of human and mouse multi-omics data; mechanisms of gene expression, RNA processing, and mRNA translation in neural systems.

EDUCATION

New York University, New York, NY

Doctor of Philosophy, Basic Medical Science, August 2013 – May 2020

- Advisor: Eric Klann, Ph.D.
- Dissertation: “Molecular and computational examination of *de novo* protein synthesis in fragile X syndrome”

Williams College, Williamstown, MA

Bachelor of Arts, Biology (honors) and Economics, September 2008 – June 2012

- Advisor: Tim J. Lebestky, Ph.D.
- Honors Thesis: “The role of DopR neuronal circuits in regulating endogenous arousal in *D. melanogaster*”

RESEARCH EXPERIENCE

Stanley Center for Psychiatric Research, Broad Institute, Cambridge, MA

Research Scientist II

July 2026 – present

Lead computational and experimental investigations of genetic mouse models of schizophrenia and bipolar disorder to identify convergent disease mechanisms and experimentally tractable therapeutic targets.

- Direct cross-disciplinary projects integrating synaptic proteomics, bulk and single-nucleus transcriptomics, spatial transcriptomics, behavioral phenotyping, and molecular experiments across multiple psychiatric risk-gene models.
- Lead comparative and meta-analytic studies of mouse models carrying mutations in schizophrenia and bipolar disorder risk genes, including *Akap11*, *Atp9a*, *Gria3*, *Grin2a*, *Sp4*, *Srrm2*, *Trio*, *Xpo7*, and *Zmym2*.
- Integrate mouse multi-omics datasets with human postmortem brain transcriptomic and proteomic data to identify conserved molecular pathways, candidate biomarkers, and mechanisms of synaptic dysfunction.
- Develop mechanistic hypotheses from large-scale datasets and design molecular, cellular, and biochemical experiments to test candidate disease drivers, including pathways related to neuronal activity, synapse organization, RNA processing, and mRNA translation.
- Directly supervise research associates and coordinate projects across genetics, proteomics, computational biology, neurobiology, and therapeutic-development teams.

Research Scientist I

January 2024 – June 2026

Led integrative analyses and functional studies of psychiatric disease-associated mouse models, extending discoveries from individual models into cross-model and human disease frameworks.

- Led multi-omic characterization of *Srrm2* haploinsufficient mice, integrating bulk RNA-seq, alternative-splicing analysis, single-nucleus RNA-seq, synaptic proteomics, RNAscope, electrophysiology, and behavioral phenotyping.
- Identified altered *Agap3* splicing, selective reduction of SynGAP- γ , and reduced mature oligodendrocyte abundance and myelin-associated programs in *Srrm2* mutant brains.

- Performed Stouffer’s Z-based meta-analyses of synaptic proteomic datasets across multiple schizophrenia and bipolar disorder risk-gene models, ages, and brain regions to identify convergent and model-specific molecular alterations.
- Integrated bulk RNA-seq, single-nucleus RNA-seq, and synaptic proteomics to distinguish transcriptional, cellular-compositional, and post-transcriptional contributions to altered protein abundance.
- Designed and implemented a scalable, low-cost workflow for whole-brain single-nucleus RNA sequencing, encompassing experimental sample preparation, library generation, computational processing, cell-type annotation, and differential-expression analysis.
- Directed follow-up studies of RNA splicing and translation, synaptic protein regulation, neuronal activity, primary neuronal cultures, viral perturbation strategies, and pharmacological treatment of mouse models.
- Coordinated experimental and analytical work across collaborators and mentored research associates in study design, data analysis, scientific interpretation, and presentation of results.

Relevant publication: **Aryal, S.**, Geng, C., Yin, M. A., Kwon, M. J., Farsi, Z., Goble, N., Asan, A. S., Brenner, K. S., Shepard, N., Geller, A., et al. Srrm2 haploinsufficiency drives SynGAP- γ reduction, Agap3 mis-splicing, and oligodendrocyte deficits in a genetic mouse model of schizophrenia. *Cell Reports*, 45(6), 2026.

Postdoctoral Associate, Advisor: Morgan Sheng, M.D., Ph.D. October 2020 – December 2023
Investigated molecular and cellular alterations in human schizophrenia and bipolar disorder brain tissue and in genetic mouse models using proteomic, transcriptomic, single-cell, and computational approaches.

- Conducted differential and weighted gene co-expression network analyses (WGCNA) of deep human postmortem synaptic proteomic datasets from schizophrenia and bipolar disorder, identifying coordinated reductions in synaptic protein networks and increases in autophagy- and protein-homeostasis-associated signatures.
- Led comparison of human synaptic proteomic alterations with purified-synapse proteomics from genetic mouse models to identify conserved disease-associated pathways and experimentally tractable mechanisms.
- Developed computational workflows for quality control, differential abundance testing, network analysis, pathway enrichment, cross-species comparison, and visualization of large-scale proteomic datasets.
- Contributed integrative proteomic and transcriptomic analyses to the characterization of *Grin2a* and other psychiatric risk-gene mouse models and connected molecular changes with cellular, circuit, and behavioral phenotypes.
- Collaborated across proteomics, neurobiology, genetics, and computational teams to translate human postmortem findings into hypotheses that could be tested in model systems.

Relevant publications:

- **Aryal, S.**, Bonanno, K., Song, B. J., Mani, D. R., Keshishian, H., Carr, S. A., Sheng, M., Dejanovic, B. Deep proteomics identifies shared molecular pathway alterations in synapses of patients with schizophrenia and bipolar disorder and mouse model. *Cell Reports*, 42(5), 2023.
- Farsi, Z., Nicolella, A., Simmons, S. K., **Aryal, S.**, Shepard, N., Brenner, K., Lin, S., Herzog, L. E., Moran, S. P., Stalnaker, K. J., et al. Brain-region-specific changes in neurons and glia and dysregulation of dopamine signaling in *Grin2a* mutant mice. *Neuron*, 111(21):3378–3396.e9, 2023.

Center for Neural Science, New York University, New York, NY

Graduate Research Assistant, Advisor: Eric Klann, Ph.D. June 2014 – June 2020
Investigated dysregulated mRNA translation in fragile X syndrome using genome-wide, biochemical, molecular, and computational approaches.

- Designed and performed ribosome profiling and TRAP-seq experiments in fragile X syndrome model mice and primary neurons to measure steady-state and activity-dependent changes in

translation.

- Performed all computational analyses of high-throughput sequencing data, including read processing, transcript-level quantification, differential translation analysis, coding-sequence feature analysis, pathway enrichment, and data visualization.
- Developed an experimental and computational approach to estimate ribosome translocation rates in primary neurons and used it to identify coding-sequence length-dependent translational dysregulation.
- Demonstrated that genetic removal of p70 S6K1 corrected translation abnormalities associated with loss of FMRP, linking altered signaling to specific features of translational control.
- Conducted primary neuronal culture, biochemical fractionation, metabolic labeling, immunoprecipitation, western blotting, qPCR, and molecular validation of genome-wide findings.
- Contributed molecular and translational analyses to work demonstrating that cell-type-specific disruption of cortico-striatal circuitry drives repetitive behavior in fragile X syndrome model mice.

Relevant publications:

- **Aryal, S.**, Longo, F., Klann, E. Genetic removal of p70 S6K1 corrects coding sequence length-dependent alterations in mRNA translation in fragile X syndrome mice. *Proceedings of the National Academy of Sciences USA*, 118(18):e2001681118, 2021.
- Longo, F., **Aryal, S.**, Anastasiades, P. G., Maltese, M., Baimel, C., Albanese, F., Tabor, J., Zhu, J. D., Oliveira, M. M., Gastaldo, D., et al. Cell-type-specific disruption of cortico-striatal circuitry drives repetitive patterns of behavior in fragile X syndrome model mice. *Cell Reports*, 42(8), 2023.
- **Aryal, S.**, Klann, E. Turning up translation in fragile X syndrome. *Science*, 361(6403):648–649, 2018.

*Neuroscience Research Partnership, A*STAR - Duke University - NUS, Singapore*

Research Assistant, Advisor: Adam Claridge-Chang, Ph.D.

August 2012 – May 2013

Developed quantitative approaches to analyze behavior and communicate effect sizes in *D. melanogaster* neurogenetics studies.

- Designed and implemented MATLAB pipelines to process and analyze high-throughput fly motion-tracking data, quantify defensive and anxiety-related behaviors, and compare behavioral effects across experimental groups.
- Developed a modular software library for estimation graphics that emphasizes effect sizes and confidence intervals rather than reliance on p-values alone.
- Contributed behavioral and computational analyses to studies of conserved neural pathways regulating defensive behavior.
- Awarded the Singapore International Pre-Graduate Award (SIPGA) for this work.

Relevant publications:

- Ho, J., Tumkaya, T., **Aryal, S.**, Choi, H., Claridge-Chang, A. Moving beyond P values: data analysis with estimation graphics. *Nature Methods*, 16(7):565–566, 2019.
- Mohammad, F., **Aryal, S.**, Ho, J., Stewart, J. C., Norman, N. A., Tan, T. L., Eisaka, A., Claridge-Chang, A. Ancient anxiety pathways influence *Drosophila* defense behaviors. *Current Biology*, 26(7):981–986, 2016.

Department of Biology, Williams College, Williamstown, MA

Honors Student, Advisor: Tim J. Lebestky, Ph.D.

September 2011 – May 2012

Investigated neuronal circuits regulating arousal and their connectivity in the *D. melanogaster* brain by conditionally activating neurons expressing temperature-gated trpA1 channels, followed by behavioral analysis using sleep and locomotor assays.

Published Articles

1. Song, B. J., Ge, Y., Nicoletta, A., Kwon, M. J., Lodder, B., Bonanno, K., Valle-Tojeiro, A., Hartley, N. D., Perzel Mandell, K., Adeleye, J., et al. Elevated synaptic PKA activity and

ADDITIONAL
PUBLICATIONS

abnormal striatal dopamine signaling in Akap11 mutant mice, a genetic model of schizophrenia and bipolar disorder. *Nature Communications*, 16(1):10793, 2025.

2. Perzel Mandell, K. A., Simmons, S. K., Nadig, A., Farsi, Z., Huang, W.-C., **Aryal, S.**, Kwon, M. J., Song, B., Brenner, K., Shepard, N., et al. Meta-analysis of the brain transcriptomes of multiple genetic mouse models of schizophrenia highlights dysregulation in striatum and thalamus. *Translational Psychiatry*, 15(1):345, 2025.
3. Longo, F., Mancini, M., Ibraheem, P. L., **Aryal, S.**, Mesini, C., Patel, J. C., Penhos, E., Rahman, N., Mamcarz, M., Santini, E., et al. Cell-type-specific disruption of PERK-eIF2 α signaling in dopaminergic neurons alters motor and cognitive function. *Molecular Psychiatry*, 26(11):6427–6450, 2021.

Preprints

1. Wu, M., Huang, J., **Aryal, S.**, Farsi, Z., Chen, H., Zhou, Y., Luo, S., Wang, W. X., Bonanno, K., Yin, M., et al. Spatially resolved translational dysregulation in Grin2a^{+/-} mouse model of schizophrenia. *bioRxiv*, 2025.
2. Huang, W.-C., Mandell, K. P., **Aryal, S.**, Song, B. J., Valle-Tojeiro, A., Goble, N., Geng, C., Asan, A. S., Liu, X.-M., Dennis, C., et al. Epigenetic changes, neuronal dysregulation and metabolomic abnormalities in Zmym2 mutant mice, a genetic model of schizophrenia and neurodevelopmental disorders. *bioRxiv*, 2025.
3. Kwon, M. J., Perzel Mandell, K. A., **Aryal, S.**, Geng, C., Nicoletta, A. A., Valle-Tojeiro, A., Natarajan, S., Misri, D., Hall, A., Song, B. J., et al. Multi-omics characterization reveals brain-wide disruption of synapses and region- and age-specific changes in neurons and glia in Sp4 mutant mice, a genetic model of schizophrenia and bipolar disorder. *bioRxiv*, 2024.
4. Falkovich, R., **Aryal, S.**, Wang, J., Sheng, M., Bathe, M. Synaptic composition, activity, mRNA translation and dynamics in combined single-synapse profiling using multimodal imaging. *bioRxiv*, 2024.
5. Huang, W.-C., Kast, R., Mandell, K. P., Dejanovic, B., Bonanno, K., **Aryal, S.**, Farsi, Z., Wilde, J., Wang, D., Gao, X., et al. Brain-region-specific changes and dysregulation of activity regulated genes in Gria3 mutant mice, a genetic animal model of schizophrenia. *bioRxiv*, 2024.

HONORS AND AWARDS

- Awarded *registration and travel scholarship* to attend the 2017 Summer Institute in Statistics for Big Data, University of Washington, Seattle. April 2017
- Inducted into the *Sigma Xi Scientific Research Society*. June 2012
- Recipient of the *Robert L. Gaudino Fellowship* at Williams College to conduct independent Economics research on healthcare access in rural Nepal. January 2010
- Awarded *Brilliance in Nepal* by the British Council, Kathmandu, for achieving the highest marks in University of Cambridge Advanced-Level Biology examinations across Nepal, May/June 2007 session. April 2008
- Graduated *Valedictorian* from Budhanilkantha School, the national school of Nepal. May 2007

SELECTED PRESENTATIONS

- *Invited speaker*, Stanley Center Scientific Advisory Board Meeting, Broad Institute of MIT and Harvard. Presented work on molecular, cellular, and functional consequences of *Srrm2* haploinsufficiency. 2024
- *Invited speaker*, Nanosymposium on “Neurodevelopmental Disorders – Mechanisms,” Society for Neuroscience Annual Meeting. 2017
- *Invited speaker*, Annual Molecular Pharmacology Retreat, Sackler Institute at NYU School of Medicine. 2017

MENTORING AND TEACHING

Directly supervise research associates and mentor undergraduate researchers in experimental design, computational analysis, scientific interpretation, project management, and presentation of results.

<i>Name</i>	<i>Role or program</i>	<i>Undergraduate institution</i>	<i>Dates</i>
Margaret A. Yin	Research Associate; direct report	Harvard College	August 2025 – present
Chuhan Geng	Research Associate; direct report	Williams College	July 2023 – July 2025
Arianna Antigua	Broad Summer Research Program	Brown University	Summer 2024
Devorah Wertheimer	MIT UROP	Massachusetts Institute of Technology	September 2024 – June 2025
Lainie Beauchemin	MIT UROP	Massachusetts Institute of Technology	November 2020 – May 2022

- Graduate Teaching Assistant to Dr. Angus Wilson for “Introduction to Research,” Sackler Institute at NYU School of Medicine. *Fall 2017*

ADDITIONAL PROFESSIONAL EXPERIENCE

- *Intern*, Office of Industrial Liaison, NYU Langone Health. Developed a business plan for an NYU technology by assessing its scientific basis, competitive advantages, market opportunity, and potential business model. *July – November 2017*

SKILLS

- Programming languages: Python, R, MATLAB, Shell scripting (Bash).
- Applications: Git, L^AT_EX, Jupyter notebook.
- Environments: High Performance Computing (SGE/Slurm), Unix/Linux, Windows.
- Computational approaches: Multi-omics integration, network analysis, meta-analysis, differential expression, alternative-splicing analysis, cell-type annotation, and analysis of bulk and single-cell sequencing data.
- Assays: Transcriptomics (single-nucleus and bulk RNA-Seq), translatoomics (ribosome profiling, Trap-seq, and RiboMap), and proteomics (MS/MS of purified synapses).
- Laboratory techniques: Primary neuron culture, transgene delivery with AAV and lentiviral vectors, in vivo pharmacology in mice, nucleic acid purification, high-throughput library preparation, qPCR, immunoprecipitation, western blotting, immunofluorescence, confocal microscopy, and immunological assay development.